Important questions

Academic year:2025-26

Course name: Programming with Python

Course code: 2501IT42

	Unit-1
1	Write about the following terms. i. Keywords ii. Identifiers iii. Data types iv. Variables v. Comments vi. quotations
2	List and explain various operators in python with suitable examples.
3	List and explain the features of python.
4	With syntax and program explain various if statements (if, if-else, if-elif-else).
5	Explain for statement with sequence and range function along with example.
6	Explain loop control statements (break, continue, pass) with necessary programs
7	Explain the working of else with loops with examples.
8	Explain the concept of nested loop with examples(pattern programs)
9	Explain top-down and bottom-up approaches.
10	Programs(even or odd, factorial, Fibonacci, multiplication table, reverse of a number, palindrome, sum of individual digits, etc.,)
	Unit-2
1	List and explain the operations of list with suitable examples.
2	List and explain the operations of tuple with suitable examples.
3	List and explain the operations of string with suitable examples.
4	List and explain the operations of dictionary with suitable examples.
5	List and explain the operations of set with suitable examples.
6	Explain different types of comprehensions with examples.
7	Programs
	Unit-3
1	Define functions? Explain the creation and calling of a function with suitable examples.
2	Explain the different function categories with examples(like function with arguments with return statement).
3	List and explain various function arguments with suitable examples.
4	Can a function returns multiple type of data? Justify with suitable example.
5	Define recursion? Explain with suitable program.
6	Define module? Explain different built-in modules with examples.
7	Explain import, from and as keywords with program.

8	How to create the package? Explain with an example.
9	Programs.
	Unit-4
1	Explain object-oriented principles with examples.
2	Define class and object? How to create the class and object? How to access the members of class using the object reference?
3	Write a short notes on following terms. i. Constructor ii. Method iii. Destructor
4	Explain different types of methods used in classes? How to call those methods?
5	Define variable? Explain different types of variables in classes? How to access those variables?
6	Define Inheritance? Explain different types of inheritances supported by python.
7	Explain following terms. i. Operator overloading ii. Method overloading and overriding iii. Constructor overloading and overriding.
8	Explain super() function with example.
9	Explain the file handling functions with examples.
10	Explain file opening modes with examples
11	Explain log files and config files with examples
12	Programs.
	Unit-5
1	Explain various types of errors with examples.
2	Define exception? Explain different exception handling techniques.
3	Can we write the try with multiple except statements? Justify your answer with examples.
4	Explain different built-in exceptions in python with examples.
5	How to create and raise the user defined exceptions with an examples.
6	Explain different clean-up activities with suitable examples.
7	Explain how to create the GUI application in python with examples?
8	Explain different widgets used in GUI based applications with examples.
9	Write the differences between terminal based and GUI based programming.
10	Programs